Edge Intelligence Laboratory

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25MML0007

1. cv2.CAP_DSHOW it is attribute for accessing the camera. It helps Opencv connect to the webcam

2. cv2.VideoCapture - Creates a video capture object to access a camera or video file.

[1]:
```
import cv2
cap=cv2.VideoCapture(0,cv2.CAP_DSHOW)
print("camera opened:",cap.isOpened())
```

```
camera opened: True
```

[2]:
```
import cv2
cap=cv2.VideoCapture(0)

while True:
    ret, frame=cap.read()

    if not ret:
        print("Failed to grab frame")
        break

    cv2.imshow("Laptop Camera",frame)

    if cv2.waitKey(1)&0XFF==ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```

3. cap.isOpened() – Checks whether the camera is opened

4. cv2.imshow(“Laptop Camera”,frame)- Opens a window titled "Laptop Camera". Displays the current video frame.

5. One video file of format .mp4 file is taken.

6. Extracts one frame from the video

7. Prints image size

8. Prints pixel value at (100,100)

9. Shows original frame Vs grayscale frame

```
]: import cv2

cap = cv2.VideoCapture(r"C:\Users\oviya\Downloads\026c7465-309f6d33.mp4")

ret, frame = cap.read()

if ret:
    print("Frame shape:", frame.shape)
    print("Pixel value at (100,100):", frame[100,100])

    cv2.imshow("Original Frame", frame)

    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
    cv2.imshow("Grayscale Frame", gray)

    cv2.waitKey(0)
else:
    print("Failed to read frame")

cap.release()
cv2.destroyAllWindows()
```

```
Frame shape: (720, 1280, 3)
Pixel value at (100,100): [71 76 87]
```

Original Frame

Grayscale Frame

10. Canny Edge detection

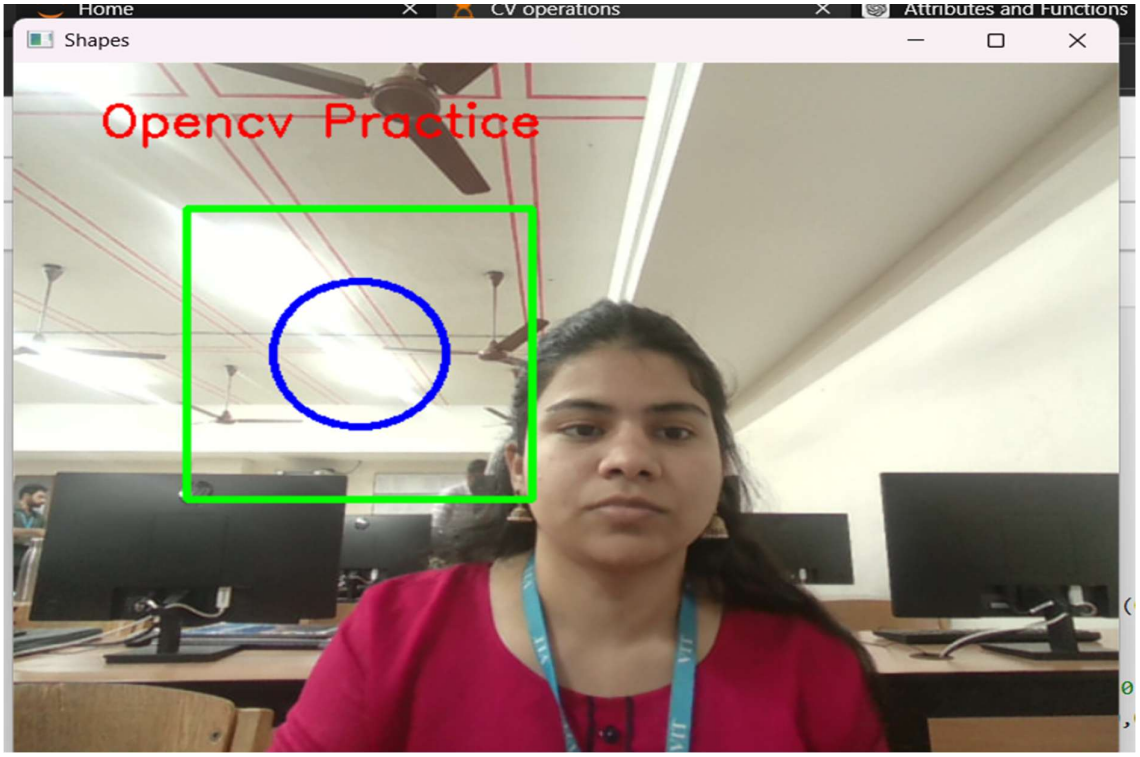
Edge detection helps to:

- Identify object boundaries
- Detect shapes
- Reduce image data while keeping important structure

```
[3]: import cv2

     cap=cv2.VideoCapture(0)

     while True:
         ret, frame=cap.read()
         if not ret:
             break
         edges=cv2.Canny(frame,100,200)

         cv2.imshow("Edges",edges)
         if cv2.waitKey(1)==ord('q'):
             break
     cap.release()
     cv2.destroyAllWindows()
```

- It Shows white edges on a black background.

Edges
VIT
HOSTELLER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11. Negative Image

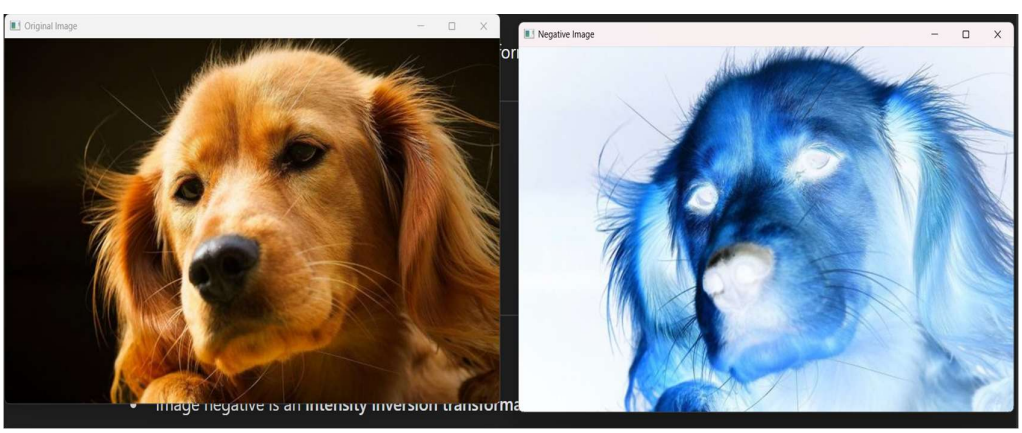
For an 8-bit image, the mathematical formula is:

$$\text{Negative Pixel} = 255 - \text{Original Pixel}$$

- So dark becomes light and light becomes dark.

```
[*]: import cv2

# Read image
img = cv2.imread(r"C:\Users\oviya\OneDrive\Pictures\wp.jpg")


# Convert to negative
negative = 255 - img

# Display images
cv2.imshow("Original Image", img)
cv2.imshow("Negative Image", negative)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

Negative Image

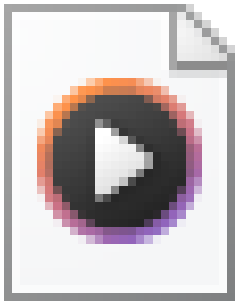


12.Motion Detection:

- Captures two consecutive frames from the webcam.
- Computes absolute difference between frames.
- Converts the difference to grayscale.
- Applies Gaussian blur to remove noise.
- Uses thresholding to highlight moving areas.

[14]:
```
import cv2

cap = cv2.VideoCapture(0)

ret, frame1 = cap.read()
ret, frame2 = cap.read()

while True:
    diff = cv2.absdiff(frame1, frame2)

    gray = cv2.cvtColor(diff, cv2.COLOR_BGR2GRAY)
    blur = cv2.GaussianBlur(gray, (5, 5), 0)

    _, thresh = cv2.threshold(blur, 20, 255, cv2.THRESH_BINARY)

    cv2.imshow("Motion", thresh)
#difference = frame(t) - frame(t+1)
    frame1 = frame2
    ret, frame2 = cap.read()

    if not ret:
        break

    if cv2.waitKey(1) & 0xFF == ord("q"):
        break

cap.release()
cv2.destroyAllWindows()
```

Motion Video:



20260305-1153-51.3671259.mp4